

ious. Thus far, it has developed an autonomous control system called 'Selenium,' which is basically a vehicle agnostic operating system applicable to anything from forklifts and cargo pods to vehicles. It also offers a cloud-based fleet management system to schedule and coordinate a fleet of autonomous vehicles, enabling smartphone booking, route optimisation and data exchange between vehicles without human intervention.

SoCs optimised for neural networks

Chip vendors are using everything from CPUs and GPUs to FPGAs and DSPs to enable CNN on vision SoCs. Currently, CNN-based architectures are mainly mapped on CPU/GPU architectures, which are not suitable for low-power and low-cost embedded products. But this is changing with new embedded vision processors introduced into

the market. These vision processors implement CNN on a dedicated programmable multicore engine that is optimised for efficient execution of convolutions and the associated data movement. Such an engine is organised to optimise dataflow and performance, using a systolic array of cores called processing elements (PEs). One PE passes its results directly to the other using FIFO buffers, without storing and loading the data to the main memory first, which effectively eliminates shared memory bottlenecks.

Vision processors offer flexibility on how PEs are interconnected, allowing designers to easily create different application-specific processing chains by altering the number and interconnection of cores used to process each CNN stage or layer. This new class of embedded vision processor is not restricted to just object detection with CNN. Other algorithms like

histogram of oriented gradients, Viola-Jones and SIFT can be implemented as well, giving designers the flexibility to tailor the processor to their application.

In parallel processing, an image to which certain parameters are applied produces another image, and as another filter is applied to the image, it produces another image. But the good news is that there are tricks to simplify the process and remove unwanted connections.

The challenge, however, remains in handling a number of different nodes in CNNs, as developers can't predetermine which node needs to be connected to another node. That's why they need a programmable architecture and can't hardwire the connections. Rather than designing a different SoC architecture or optimising it every time new algorithms pop up, a CNN processor only needs a fairly simple algorithm that comes with fewer variables. **EFY**

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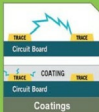
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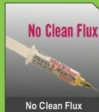
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